

palimpsestr: User Manual

Complete Reference Guide v0.18.0

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1 Introduction

palimpsestr is an R package for the probabilistic decomposition of archaeological palimpsests using the Stratigraphic Entanglement Field (SEF) framework. It integrates spatial proximity, vertical distribution, chronological overlap, and cultural similarity to estimate latent depositional phases.

1.1 Installation

```
# From GitHub
devtools::install_github("enzococca/palimpsestr")

# From local source
install.packages("palimpsestr_0.18.0.tar.gz", repos = NULL, type = "source")
```

1.2 Required and optional packages

	Package	Status	Purpose
stats, utils, graphics, grDevices	Required	Core R	
ggplot2	Suggested	Publication-quality plots (<code>gg_*</code> functions)	
plotly	Suggested	Interactive visualization (<code>as_plotly()</code>)	
sf	Suggested	GIS export (<code>as_sf_phase()</code> , <code>as_sf_links()</code>)	
DBI, RSQLite, RPostgres	Suggested	Database import (<code>read_db()</code>)	

2 Data Preparation

2.1 Required columns

Every dataset must contain:

Column	Type	Description
x	numeric	Easting coordinate (metres)
y	numeric	Northing coordinate (metres)
z	numeric	Depth or elevation (metres)
date_min	numeric	Start of chronological interval
date_max	numeric	End of chronological interval
class	character	Material class (ceramic, lithic, bone, metal, etc.)

2.2 Optional columns

	Column	Type	Description
	id	any	Unique identifier for each find
	context	character	Stratigraphic unit label (e.g., “SU_1”)
	taf_score	numeric (0–1)	Taphonomic disturbance score
	true_phase	integer	Known phase (for validation only)

2.3 Notes on Z coordinates

Each artifact needs its own z value. If you don’t have point-level depth measurements, you can assign the mean depth of the stratigraphic unit (US) to all finds within it:

```
# Example: assign US mean depth to finds
data$z <- ave(data$z_us, data$context, FUN = mean)
```

2.4 Simulating test data

```
# Well-separated phases (low mixing)
easy <- archaeo_sim(n = 150, k = 3, seed = 1, mixing = 0.05)

# Moderate palimpsest
moderate <- archaeo_sim(n = 200, k = 3, seed = 1, mixing = 0.30)

# Compressed palimpsest (hard to resolve)
compressed <- archaeo_sim(n = 250, k = 4, seed = 1, mixing = 0.50)
```

Parameters:

- **n**: number of artifacts
- **k**: number of phases
- **seed**: random seed for reproducibility
- **mixing**: proportion of inter-phase mixing (0 = none, 1 = complete)

2.5 Built-in datasets

The package includes four built-in datasets:

Dataset	Finds	Phases	Mixing	Description
demo_easy	150	3	5%	Well-separated phases
demo_moderate	200	3	30%	Moderate palimpsest
demo_compressed	250	4	50%	Heavily disturbed
villa_romana	615	–	Real data	Poggio Gramignano Roman villa (VRPG 3004)

```
data(villa_romana)
str(villa_romana)
#> 'data.frame':    615 obs. of  9 variables:
#>  $ id      : chr  "VRPG_0001" "VRPG_0002" "VRPG_0003" "VRPG_0004" ...
#>  $ x       : num  2299041 2299041 2299041 2299041 2299041 ...
#>  $ y       : num  4714126 4714126 4714126 4714126 4714126 ...
#>  $ z       : num  284 284 284 284 284 ...
#>  $ context  : chr  "US_102" "US_102" "US_102" "US_102" ...
#>  $ date_min : num  425 425 425 425 425 425 425 425 425 425 ...
#>  $ date_max : num  450 450 450 450 450 450 450 450 450 450 ...
#>  $ class    : chr  "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" ...
#>  $ taf_score: num  0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 ...
table(villa_romana$class)
#>
#>      Campionature      Intonaco Dipinto Materiale Da Costruzione
#>              5              1              38
#>      Materiale Lapideo      Pietre Dure-Gemme      Reperti Antropologici
#>              1              1              1
#>      Reperti Archeozoologici      Reperto Anforaceo      Reperto Ceramico
#>              7              76              417
#>      Reperto Faunistico      Reperto Lapideo      Reperto Metallico
#>              1              6              19
#>      Reperto Numismatico      Reperto Vitreo      Resti Di Pavimentazioni
#>             10             25              1
#>      Scarti Di Produzione      Stucco
#>              5              1
# Core Analysis

## Fitting the SEF model: `fit_sef()`

fit <- fit_sef(
  data      = moderate,
  coords    = c("x", "y", "z"),          # coordinate columns
  chrono    = c("date_min", "date_max"), # chronological range columns
  class     = "class",                   # material class column
  tafonomy  = "taf_score",                # optional: taphonomic score
  context   = "context",                  # optional: stratigraphic unit
)
```

```

harris = NULL,          # optional: Harris Matrix penalty
k       = 3,            # number of phases
weights = c(ws = 1, wz = 1, wt = 1, wc = 1), # domain weights
seed    = 42,           # random seed
em_iter = 100,          # max EM iterations
em_tol  = 1e-5,         # convergence tolerance
n_init  = 5             # number of random initialisations
)
print(fit)
#> <sef_fit>
#> Observations: 200
#> Estimated phases: 3
#> Mean entropy: 0.000
#> Mean local SEI: 249.857
#> Mean energy: 2.008
#> LogLik: -1177.707
#> BIC: 2572.646
#> ICL: 2572.643
#> PDI: 1.000
#> Converged: yes
#> Class model: multinomial
#> Initialisations: 5

```

2.5.1 Parameters explained

Parameter	Default	Description
k	3	Number of depositional phases to estimate
weights	c(ws=1, wz=1, wt=1, wc=1)	Weights for spatial (ws), vertical (wz), temporal (wt), and cultural (wc) evidence. Since v0.14 they scale the feature dimensions and enter the likelihood, so they can be cross-validated with <code>optimize_weights()</code>
em_iter	100	Maximum EM iterations
em_tol	1e-5	Convergence tolerance (log-likelihood change)
n_init	5	Number of random initialisations (higher = more robust, slower)
tafonomy	NULL	Column name for taphonomic disturbance (0–1)
context	NULL	Column name for stratigraphic unit labels
harris	NULL	$n \times n$ penalty matrix (from <code>harris_from_contexts()</code> or <code>read_harris()</code>)

2.5.2 Statistical-model options

These options (all optional, sensible defaults) control the statistical core:

Parameter	Default	Description
<code>class_model</code>	"multinomial"	How the material class is modelled. "multinomial" (recommended) treats it as a per-phase categorical distribution, so typology does not fragment stratigraphic units; "gaussian" is the legacy one-hot behaviour
<code>class_smoothing</code>	1	Dirichlet pseudo-count for the per-phase class probabilities (shrunk towards the global class frequencies)
<code>var_structure</code>	"diagonal"	"diagonal" (free per-dimension variances) or "spherical" (one shared variance per phase; the structure under which <code>weights</code> are identifiable)
<code>chrono_uncertainty</code>	FALSE	If TRUE, propagate each find's dating-interval width $(t_{max} - t_{min})^2/12$ into the temporal likelihood, so coarsely dated finds rely less on chronology
<code>strat_dynamic</code>	FALSE	If TRUE (with <code>context</code>), apply the stratigraphic constraint as a Neighborhood-EM field recomputed each E-step instead of a static penalty frozen at initialisation
<code>strat_beta</code>	1	Strength of the dynamic stratigraphic field
<code>noise</code>	FALSE	If TRUE, add a uniform background component (Fraley and Raftery, 1998); <code>\$noise_prob</code> becomes a genuine outlier probability and extreme finds are kept out of the phase estimates
<code>noise_prior</code>	0.05	Initial mixing weight of the noise component
<code>chrono_precision</code>	FALSE	Add $1/t_{span}$ as a feature (precise dates weighted higher)
<code>taf_as_feature</code>	FALSE	Add <code>taf_score</code> as an explicit feature dimension
<code>residuality</code>	FALSE	Add a per-find date-vs-context-mean mismatch feature (needs <code>context</code>)
<code>class_scale</code>	FALSE	Scale one-hot class columns by $1/\sqrt{n_{classes}}$ (only relevant with <code>class_model = "gaussian"</code>)
<code>subclass</code>	NULL	Optional sub-class column used instead of <code>class</code> for the categorical/one-hot block

`recommend_setup(data, context = "context", tafonomy = "taf_score")` inspects a dataset and suggests which of these to enable, flagging caveats such as unit-level coordinates or unit-tied chronology.

2.5.3 Output: the `sef_fit` object

The returned object contains:

Field	Type	Description
\$phase	integer vector	Dominant phase assignment per find
\$phase_prob	matrix	$n \times k$ matrix of phase membership probabilities (conditional on not being noise when <code>noise = TRUE</code>)
\$cat_prob	matrix	$k \times L$ per-phase class profile (multinomial model); see <code>phase_composition()</code>
\$noise_prob	numeric vector	Per-find outlier posterior (when <code>noise = TRUE</code>)
\$entropy	numeric vector	Shannon entropy per find
\$energy	numeric vector	ESE per find
\$local_sei	numeric vector	Local SEI per find
\$converged	logical	Whether EM converged
\$n_init	integer	Number of initialisations used
\$em_loglik	numeric vector	Log-likelihood trace across iterations
\$model_stats	list	Summary statistics (PDI, BIC, ICL, loglik, etc.)

The per-phase class composition can be read directly:

```
# Per-phase typological profile (multinomial model)
phase_composition(fit)
#>   phase      bone      ceramic      lithic      metal
#> 1      1 0.2429904 0.3573578 0.2118628 0.1877889
#> 2      2 0.1578399 0.3261697 0.3050742 0.2109162
#> 3      3 0.2041745 0.3028718 0.3629811 0.1299727
```

2.6 Phase count selection: `compare_k()`

When K is unknown, compare multiple values:

```
ck <- compare_k(moderate, k_values = 2:5, seed = 42,
                tafonomy = "taf_score", context = "context")
print(ck)
#>   k      pdi mean_entropy mean_local_sei mean_energy      loglik      bic
#> 1 2 0.9929875 4.860672e-03      249.8574      2.008381 -1426.116 2995.288
#> 2 3 0.9999945 6.095726e-06      249.8574      2.008381 -1177.707 2572.646
#> 3 4 0.9438440 7.784876e-02      249.8574      2.008381 -1156.909 2605.225
#> 4 5 0.9160600 1.350962e-01      249.8574      2.008381 -1114.002 2593.588
#>      icl tot_withinss pseudo_bic
#> 1 2993.343      603.9637      274.0225
#> 2 2572.643      400.1314      218.1699
#> 3 2574.085      341.0552      212.7117
#> 4 2539.550      292.7963      208.6899
```

The optimal K minimises BIC while maximising PDI.

2.7 Key diagnostics

```
# Palimpsest Dissolution Index (global separability, 0--1)
pdi(fit)
#> [1] 0.9999945

# Compact summary
sef_summary(fit)
```

```

#> $n
#> [1] 200
#>
#> $k
#> [1] 3
#>
#> $pdi
#> [1] 0.9999945
#>
#> $mean_entropy
#> [1] 6.095726e-06
#>
#> $mean_energy
#> [1] 2.008381
#>
#> $loglik
#> [1] -1177.707
#>
#> $bic
#> [1] 2572.646
#>
#> $phase_count
#>
#> 1 2 3
#> 67 66 67

# Phase probability table
head(as_phase_table(fit))
#>      id dominant_phase      phase1      phase2 phase3      entropy local_sei
#> 1 F001          3 1.124319e-19 2.076556e-40      1 4.905622e-18 252.1957
#> 2 F002          3 7.713722e-18 4.721978e-40      1 3.039479e-16 248.5608
#> 3 F003          3 4.164452e-21 2.368336e-44      1 1.954282e-19 239.7297
#> 4 F004          3 2.645145e-23 1.267429e-46      1 1.375124e-21 260.5879
#> 5 F005          3 2.204926e-15 5.781533e-35      1 7.663248e-14 279.3199
#> 6 F006          3 1.911759e-18 7.612349e-42      1 7.799693e-17 284.5397
#>      energy
#> 1 2.403297
#> 2 1.756757
#> 3 1.931740
#> 4 1.854746
#> 5 2.362912
#> 6 2.002368

# Full diagnostic table (input + diagnostics)
head(phase_diagnostic_table(fit))
#>      id      x      y      z context date_min date_max      class taf_score
#> 1 F001 12.36888 104.53174 11.79768 SU_4 970.6746 1011.055 lithic 0.4687699
#> 2 F002 19.12233 89.86743 19.78304 SU_3 960.5955 1080.232 lithic 0.3796706
#> 3 F003 24.19310 96.13386 18.97572 SU_7 966.4167 1030.552 bone 0.2288079
#> 4 F004 26.50473 99.62853 18.50500 SU_2 947.1761 1027.755 lithic 0.2246840
#> 5 F005 45.78809 91.97095 17.58970 SU_1 975.1793 1071.124 ceramic 0.1925518
#> 6 F006 32.65961 89.87875 17.67439 SU_4 946.7372 1055.110 ceramic 0.5558153
#> true_phase dominant_phase      phase1      phase2 phase3      entropy

```

```

#> 1      1      3 1.124319e-19 2.076556e-40      1 4.905622e-18
#> 2      1      3 7.713722e-18 4.721978e-40      1 3.039479e-16
#> 3      1      3 4.164452e-21 2.368336e-44      1 1.954282e-19
#> 4      1      3 2.645145e-23 1.267429e-46      1 1.375124e-21
#> 5      1      3 2.204926e-15 5.781533e-35      1 7.663248e-14
#> 6      1      3 1.911759e-18 7.612349e-42      1 7.799693e-17
#>   local_sei   energy
#> 1 252.1957 2.403297
#> 2 248.5608 1.756757
#> 3 239.7297 1.931740
#> 4 260.5879 1.854746
#> 5 279.3199 2.362912
#> 6 284.5397 2.002368

```

3 Intrusion Detection

`detect_intrusions()` returns, per find, an `intrusion_prob`, a direction factor (`older_than_context` / `in_context` / `younger_than_context`), and a signed `chrono_gap` (years). When the model was fitted with `noise = TRUE`, `intrusion_prob` is the noise-component posterior — a genuine probability of being an outlier; otherwise it is a heuristic composite of entropy, energy, and inverse local SEI.

```

fit_n <- fit_sef(moderate, k = 3, tafonomy = "taf_score",
                 context = "context", noise = TRUE, seed = 42)
di <- detect_intrusions(fit_n)
suspects <- di[di$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects), "\n")
#> Suspected intrusions: 24
head(suspects[order(suspects$intrusion_prob, decreasing = TRUE), ])
#>      id intrusion_prob direction chrono_gap
#> 13  F013      1.0000000 in_context      0
#> 24  F024      0.9999992 in_context      0
#> 112 F112      0.9999973 in_context      0
#> 1   F001      0.9999948 in_context      0
#> 145 F145      0.9999412 in_context      0
#> 44  F044      0.9998334 in_context      0

```

The `direction` distinguishes residual material (`older_than_context`) from possible latent features (`younger_than_context`); it is informative only when finds within a unit carry distinct chronologies (not under unit-tied dating).

4 Model Validation

When true phase labels are known:

```

# Adjusted Rand Index (-1 to 1, 1 = perfect)
adjusted_rand_index(fit, moderate$true_phase)
#> [1] 1

# Confusion matrix (auto-reordered for best diagonal match)
confusion_matrix(fit, moderate$true_phase)
#>      true
#> estimated 2 3 1
#>      1 67 0 0

```

```
#>      2  0 66  0
#>      3  0  0 67
```

5 Structured Exports

5.1 Summary per stratigraphic unit

```
us_summary_table(fit)
#>   context n_finds dominant_phase   purity mean_entropy mean_energy
#> 1  SU_1      18             3 0.4444444 6.763679e-05  1.933266
#> 2  SU_2      29             2 0.3793103 2.610162e-12  2.009269
#> 3  SU_3      20             2 0.3500000 2.617318e-11  2.160271
#> 4  SU_4      20             1 0.4500000 3.112000e-09  2.083647
#> 5  SU_5      16             1 0.3750000 2.177784e-11  2.005937
#> 6  SU_6      29             1 0.3793103 4.345181e-08  1.988188
#> 7  SU_7      23             2 0.4347826 1.735165e-09  2.039224
#> 8  SU_8      16             1 0.5625000 3.867276e-10  1.920946
#> 9  SU_9      29             3 0.4137931 1.081262e-08  1.942775
#>   mean_local_sei n_intrusions
#> 1      250.2169           3
#> 2      245.1726           3
#> 3      244.9395           2
#> 4      250.8428           0
#> 5      255.8361           0
#> 6      253.5263           0
#> 7      256.1566           0
#> 8      257.4712           0
#> 9      240.8673           3
```

Columns: context, n_finds, dominant_phase, purity, mean_entropy, mean_energy, mean_local_sei, n_intrusions.

5.2 Phase transition matrix

Shows how often finds from phase i sit directly above finds from phase j :

```
phase_transition_matrix(fit)
#>   phase1 phase2 phase3
#> phase1    28    35     3
#> phase2    33    31     2
#> phase3     6     0    61
```

5.3 Export all results to CSV

```
export_results(fit, dir = "results/", format = "csv", prefix = "site_X")
```

Creates 4 files:

- site_X_phases.csv — phase assignments and diagnostics per find
- site_X_intrusions.csv — intrusion probabilities
- site_X_us_summary.csv — per-US aggregated statistics
- site_X_model_summary.csv — model-level metrics (n, k, PDI, BIC, etc.)

6 Harris Matrix Integration

6.1 Auto-generate from depth ordering

```
H <- harris_from_contexts(moderate, z_col = "z", context_col = "context",
                          penalty_scale = 0.5)
dim(H)
#> [1] 200 200
```

6.2 Import from CSV

The CSV must have columns `from`, `to`, and optionally `weight`:

```
from,to,weight
SU_1,SU_2,1
SU_2,SU_3,1
SU_1,SU_3,0.5

H <- read_harris("harris_matrix.csv", contexts = data$context)
fit <- fit_sef(data, k = 3, harris = H, context = "context")
```

6.3 Validate phases against stratigraphy

```
validate_phases_harris(fit)
#>   upper lower upper_phase lower_phase consistent
#> 1  SU_1  SU_2           3           2      FALSE
#> 2  SU_2  SU_5           2           1      FALSE
#> 3  SU_5  SU_7           1           2       TRUE
#> 4  SU_7  SU_6           2           1      FALSE
#> 5  SU_6  SU_4           1           1       TRUE
#> 6  SU_4  SU_9           1           3       TRUE
#> 7  SU_9  SU_3           3           2      FALSE
#> 8  SU_3  SU_8           2           1      FALSE
```

The `consistent` column flags whether the dominant phase ordering respects stratigraphic depth. `FALSE` indicates an inversion that may warrant investigation.

7 Visualization

7.1 ggplot2 plots (static)

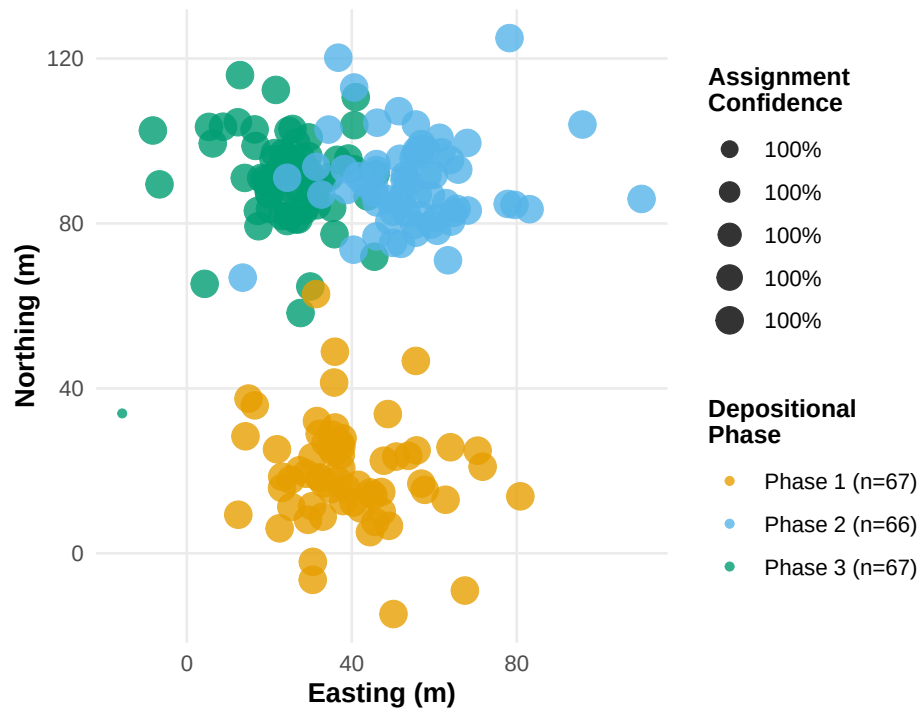
All `gg_*` functions require **ggplot2** and return ggplot objects.

```
library(ggplot2)

# Phase assignment map
gg_phasefield(fit)
```

Dominant Phase Assignment

200 finds assigned to 3 depositional phases | PDI = 1.000



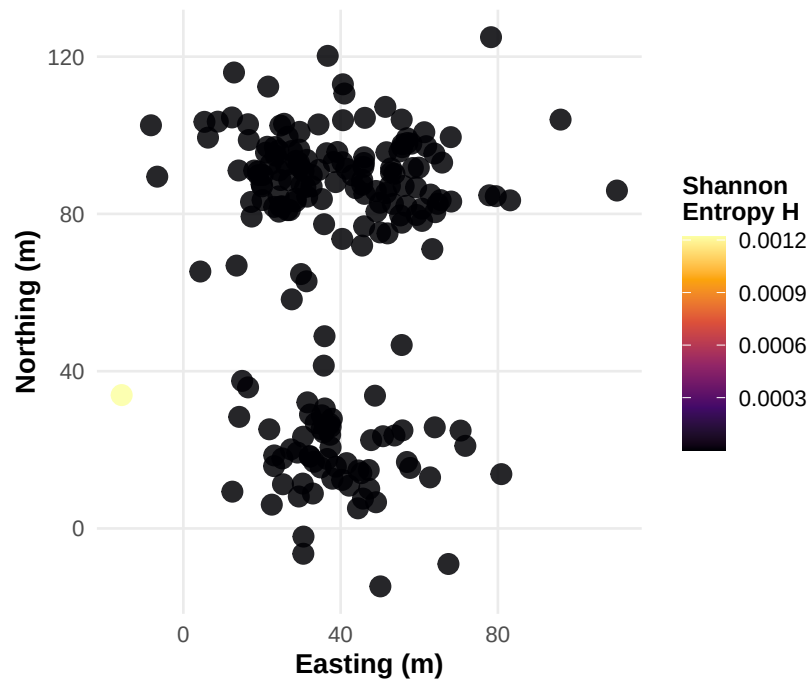
Generated by palimpsestr 0.18.0

```
# Entropy map (uncertainty)  
gg_entropy(fit)
```

Spatial Entropy Map

Entropy measures uncertainty in phase assignment

Dark = confident assignment | Bright = ambiguous (possible mixing zone)



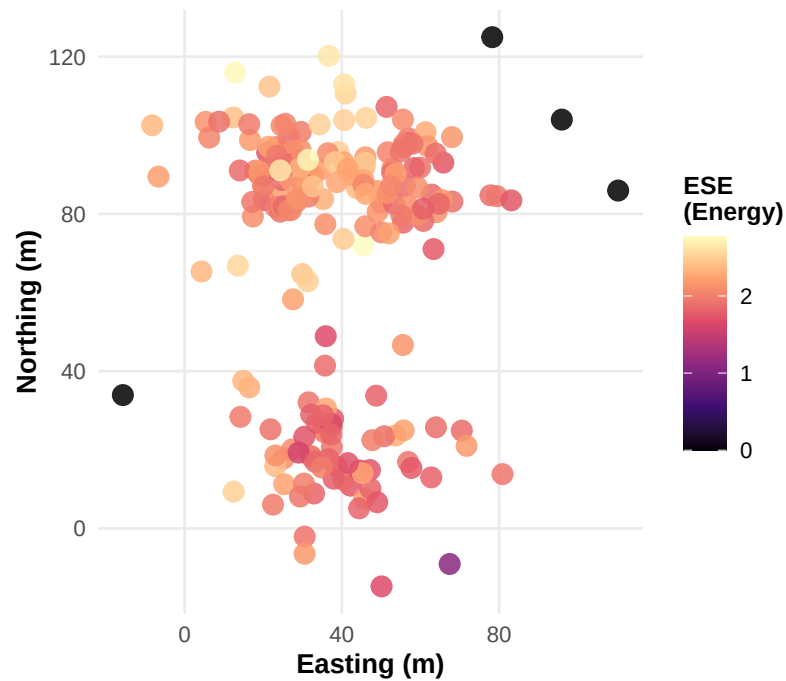
Generated by palimpsestr 0.18.0

```
# Energy map (local disruption)
gg_energy(fit)
```

Excavation Stratigraphic Energy (ESE)

Energy measures local depositional disruption

Dark = stable deposit | Bright = mixing zone (bioturbation, redeposition)

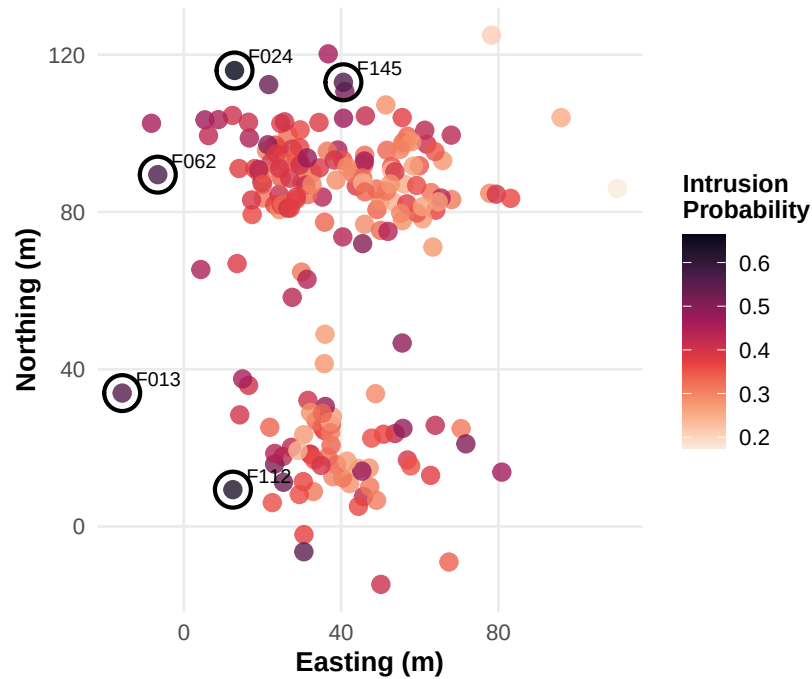


Generated by palimpsestr 0.18.0

```
# Intrusion detection map  
gg_intrusions(fit, top_n = 5)
```

Intrusion Detection Map

11 finds flagged (prob > 0.5) | Circled: top 5 suspects
Score combines high entropy + high energy + low network connectivity

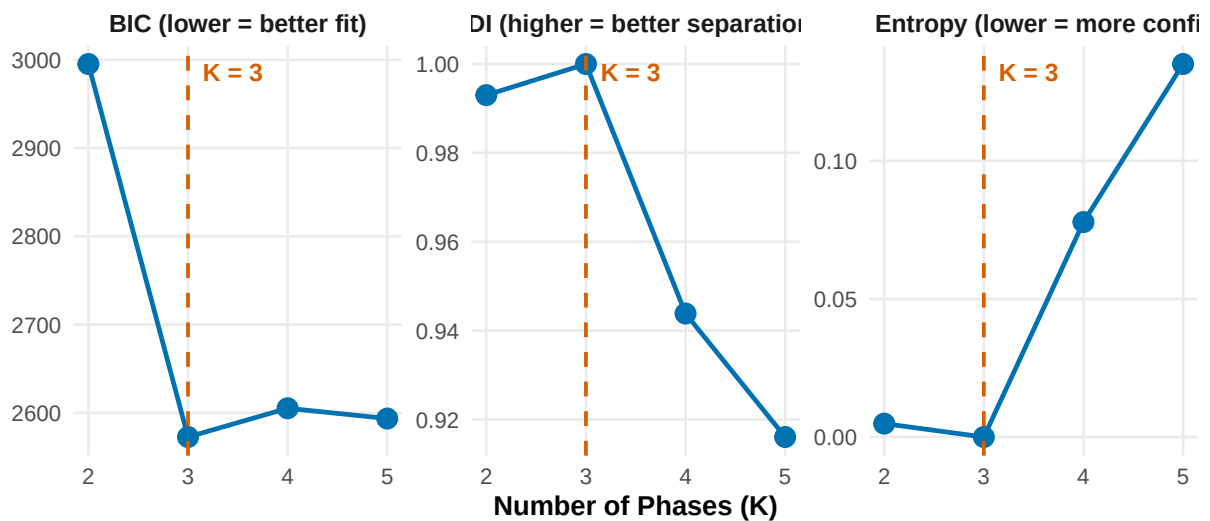


Generated by palimpsestr 0.18.0

```
# Phase count selection diagnostics
gg_compare_k(ck)
```

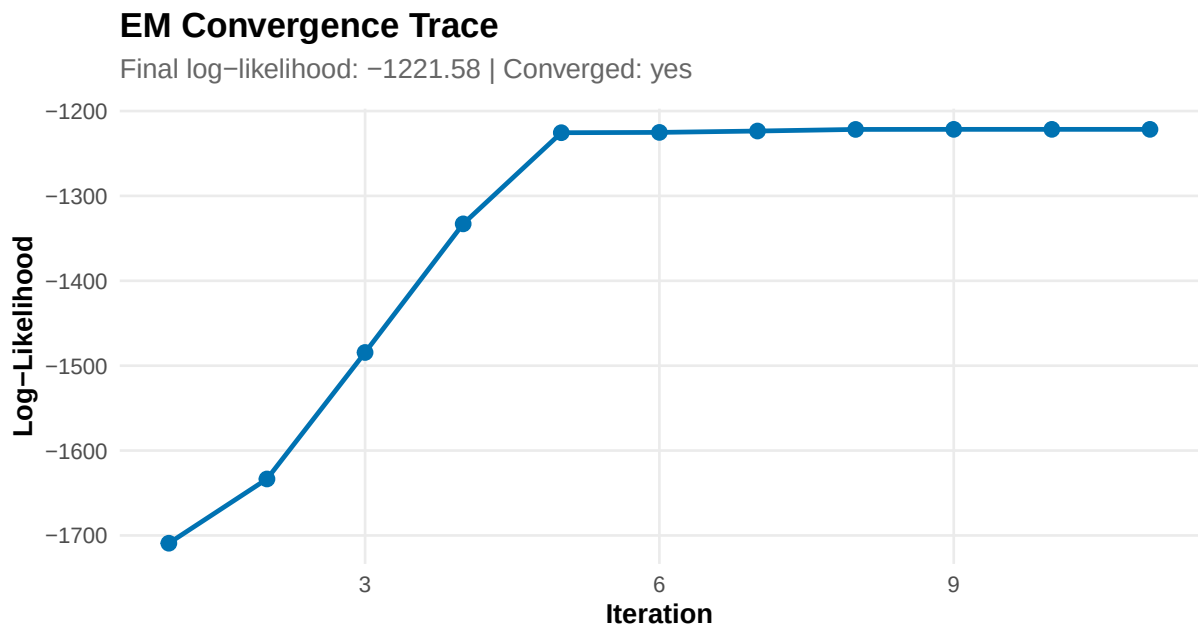
Phase Count Selection

Comparing K = 2 to 5 | Optimal K = 3 (minimum BIC)



Generated by palimpsestr 0.18.0

```
# EM convergence trace  
gg_convergence(fit)
```

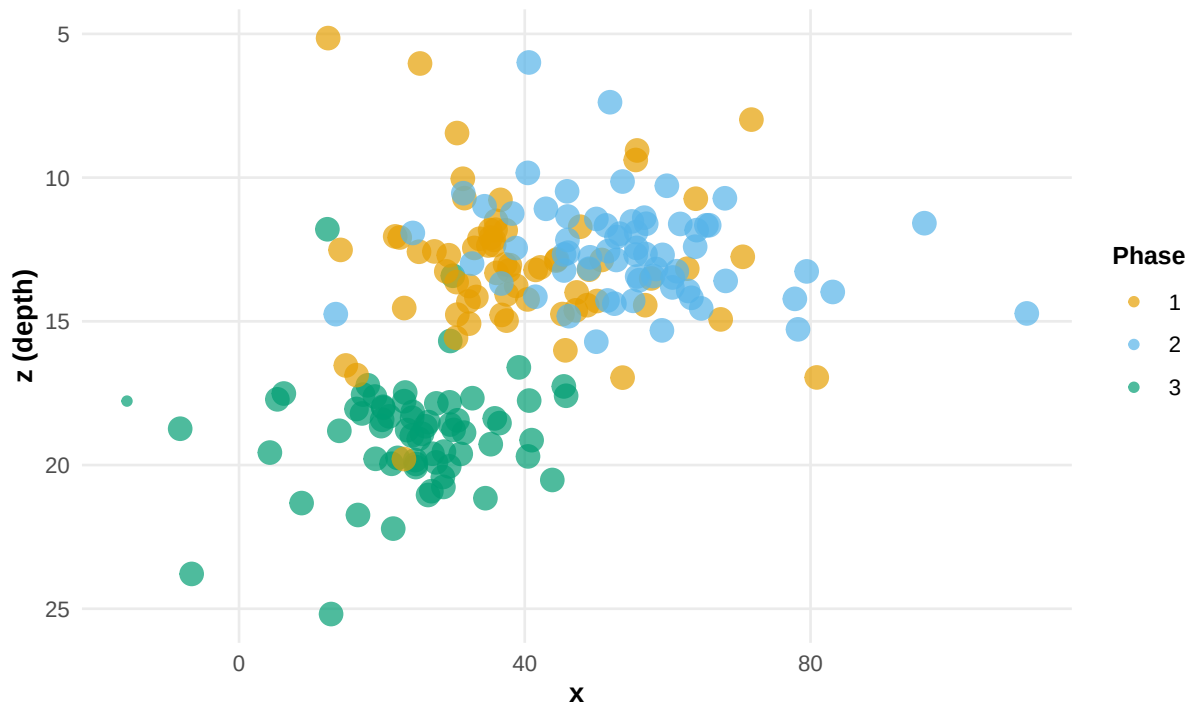


Generated by palimpsestr 0.18.0

```
# Vertical phase profile (depth vs horizontal)  
gg_phase_profile(fit)
```

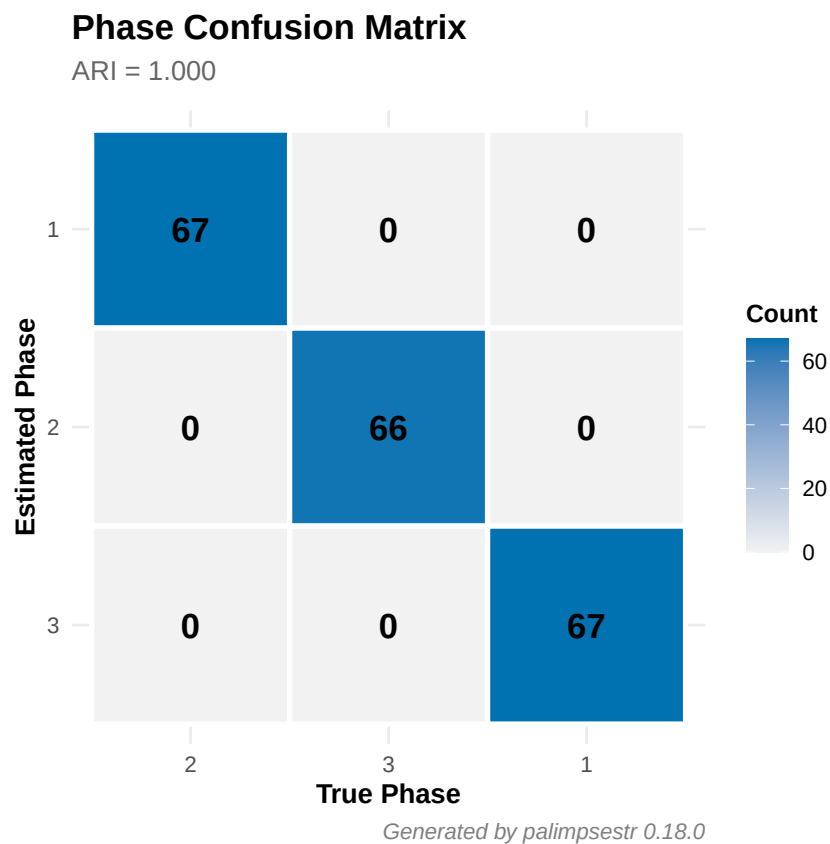
Vertical Phase Profile

Depth (z) vs horizontal position, coloured by phase



Generated by palimpsestr 0.18.0

```
# Confusion matrix heatmap  
gg_confusion(fit, moderate$true_phase)
```



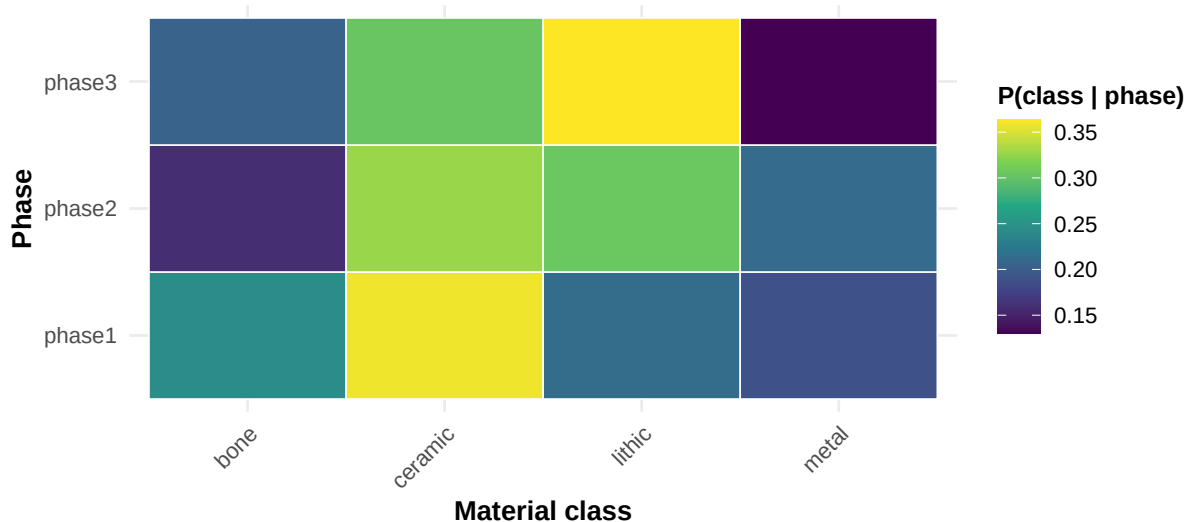
7.1.1 Model-output diagnostic plots (v0.18)

Four plots visualise the outputs of the refined statistical model:

```
# Per-phase class composition (multinomial cat_prob), heatmap by default
gg_phase_composition(fit)
```

Per-phase class composition

Estimated categorical class profile of each phase



Generated by palimpsestr 0.18.0

```
# Within-unit phase coherence; pass compare = a gaussian-class fit to contrast
gg_unit_coherence(fit)

# Non-spatial ranking of the intrusion probability (noise posterior if noise=TRUE)
gg_outliers(fit_n)

# Directional intrusions: residual vs latent (needs per-find dating)
gg_direction(fit_n)
```

7.1.2 Type longevity

`type_longevity()` aggregates the posterior to estimate the temporal envelope of each material class; `gg_longevity()` draws it as a Gantt-style chart:

```
t1 <- type_longevity(fit)
gg_longevity(t1)
```

7.2 Interactive plots (plotly)

```
library(plotly)

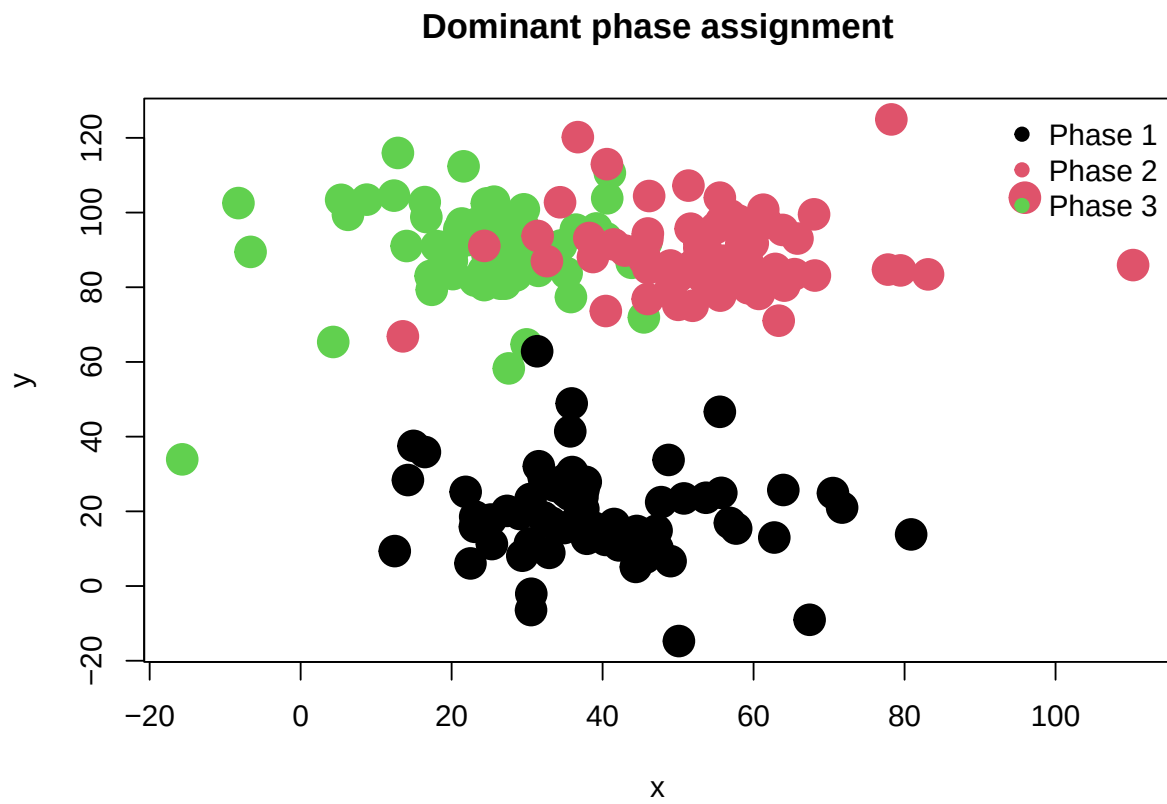
# Convert any gg_* plot to interactive
as_plotly(gg_phasefield(fit))
as_plotly(gg_entropy(fit))
as_plotly(gg_intrusions(fit))
```

Hover tooltips show: ID, context, phase probability, dating, class, entropy, energy, intrusion score.

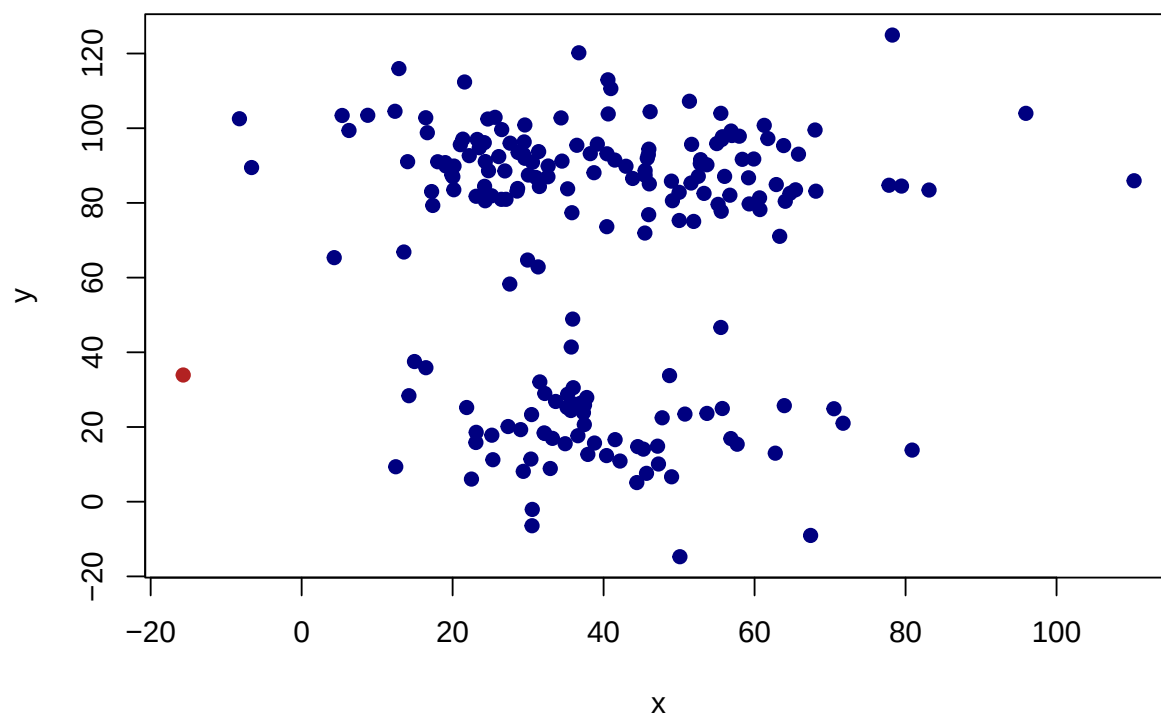
7.3 Base R plots

For environments without `ggplot2`:

```
plot_phasefield(fit)
```

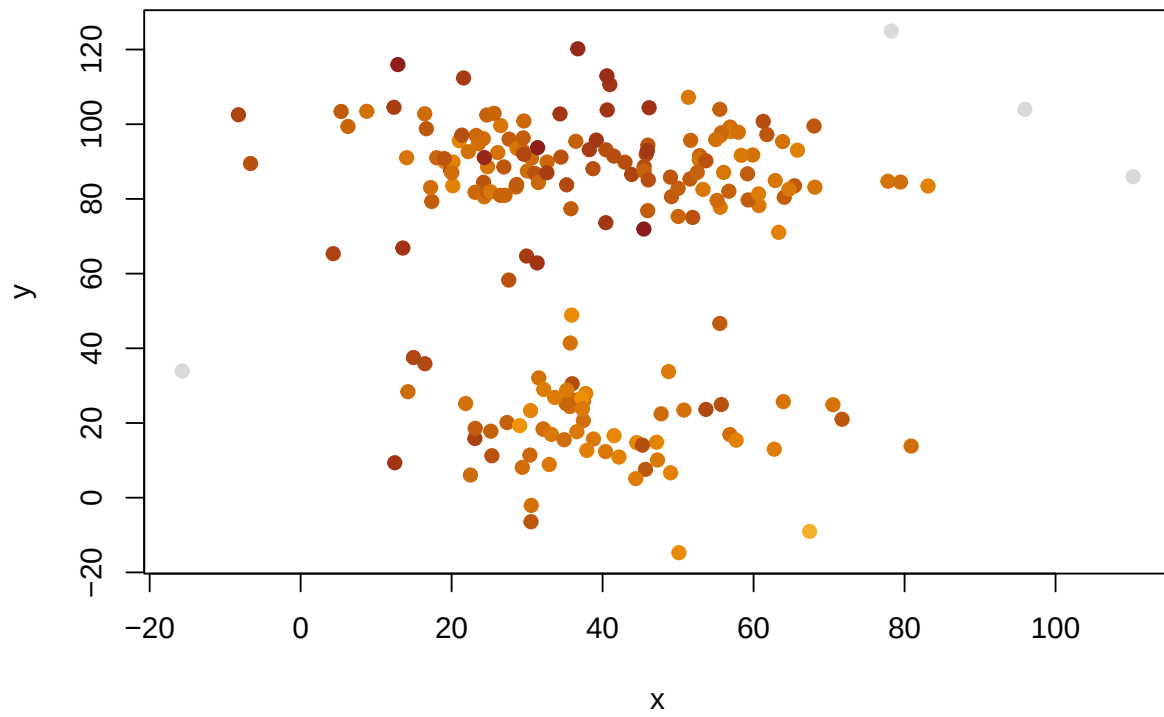


Entropy map

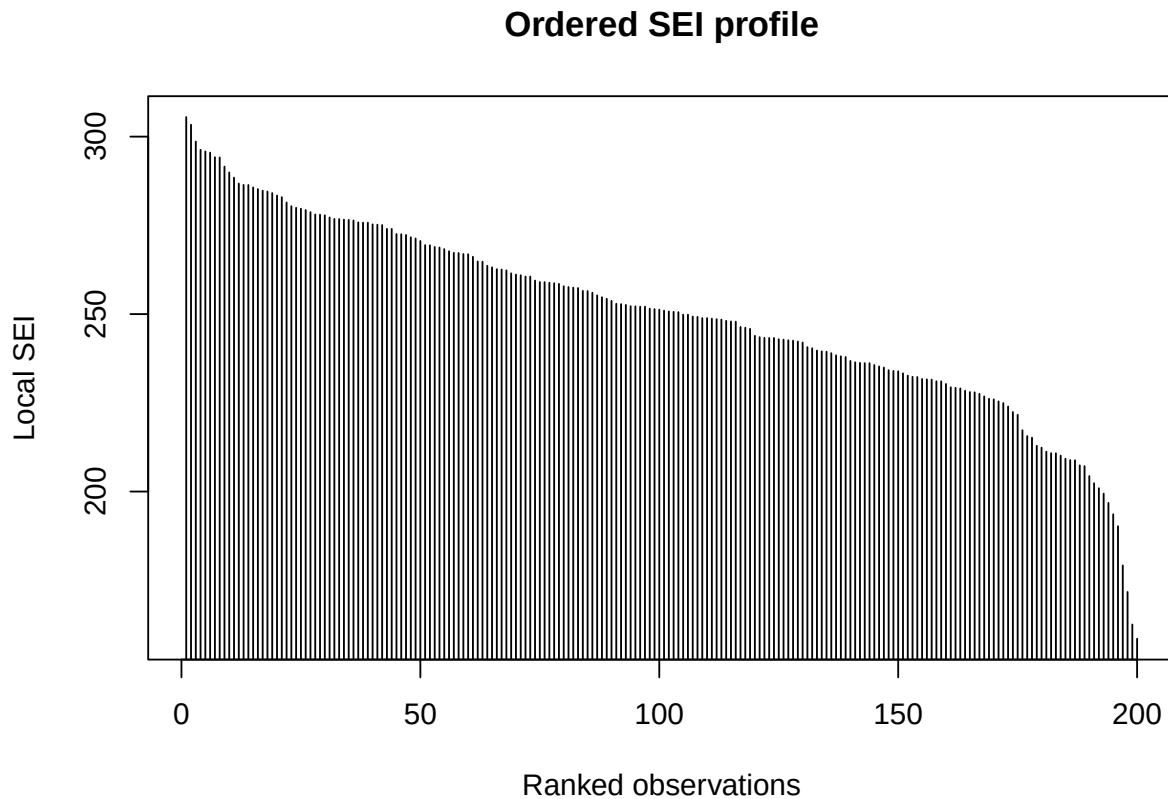


```
plot_energy(fit)
```

Excavation stratigraphic energy



```
plot_sei_profile(fit)
```



7.4 GIS export

```
library(sf)

# Point layer with phase assignments
sf_pts <- as_sf_phase(fit, crs = 32632L, dims = "XYZ")
st_write(sf_pts, "results.gpkg", layer = "phases")

# High-SEI links as lines
sf_links <- as_sf_links(fit, quantile_threshold = 0.9, crs = 32632L)
st_write(sf_links, "results.gpkg", layer = "sei_links")
```

8 Interpretive Report

Generate a plain-language report for inclusion in excavation documentation:

```
report_sef(fit, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with K = 3 phases was fitted to a dataset of 200 fi
#>
#> The Palimpsest Dissolution Index (PDI) is 1.000, indicating excellent phase separation. Mean
```

```

#>
#> Model diagnostics: LogLik = -1177.7, BIC = 2572.6.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (67 finds, 33.5%)
#>
#> - **Dating**: average 1148 CE (range: 1052 to 1246)
#> - **Material classes**: bone (17), ceramic (24), lithic (14), metal (12)
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.938
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (66 finds, 33.0%)
#>
#> - **Dating**: average 1303 CE (range: 1211 to 1393)
#> - **Material classes**: bone (10), ceramic (22), lithic (20), metal (14)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.008
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (67 finds, 33.5%)
#>
#> - **Dating**: average 1001 CE (range: 894 to 1122)
#> - **Material classes**: bone (13), ceramic (21), lithic (24), metal (9)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.080
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **11 finds** (5.5%) have an intrusion probability above 0.5, flagging them as potentially displaced
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | F024 | SU_1 | 0.663 |
#> | F112 | SU_9 | 0.625 |
#> | F062 | SU_3 | 0.569 |
#> | F145 | SU_1 | 0.560 |
#> | F013 | SU_1 | 0.556 |
#> | F120 | SU_2 | 0.547 |
#> | F012 | SU_3 | 0.531 |
#> | F090 | SU_9 | 0.530 |
#> | F048 | SU_2 | 0.519 |
#> | F080 | SU_2 | 0.506 |
#>
#> ## Stratigraphic Unit Purity
#>
#> Of **9 stratigraphic units**: **0 pure** (>90%), **0 mixed** (70-90%), **9 palimpsest** (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi

```

```

#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
#> *Report generated by palimpsestr 0.18.0*

report_sef(fit, lang = "it")
#> # Rapporto Analisi SEF
#>
#> ## Panoramica
#>
#> Il modello Stratigraphic Entanglement Field con K = 3 fasi e' stato applicato a un dataset di f
#>
#> Il Palimpsest Dissolution Index (PDI) e' 1.000, indicando una separazione eccellente delle f
#>
#> Diagnostiche: LogLik = -1177.7, BIC = 2572.6.
#>
#> ## Composizione delle Fasi
#>
#> ### Fase 1 (67 reperti, 33.5%)
#>
#> - Datazione media: 1148 d.C.
#> - Classi materiali: bone (17), ceramic (24), lithic (14), metal (12)
#> - Entropia media: 0.0000; Energia media: 1.938
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 2 (66 reperti, 33.0%)
#>
#> - Datazione media: 1303 d.C.
#> - Classi materiali: bone (10), ceramic (22), lithic (20), metal (14)
#> - Entropia media: 0.0000; Energia media: 2.008
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 3 (67 reperti, 33.5%)
#>
#> - Datazione media: 1001 d.C.
#> - Classi materiali: bone (13), ceramic (21), lithic (24), metal (9)
#> - Entropia media: 0.0000; Energia media: 2.080
#> - US: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ## Intrusioni
#>
#> 11 reperti (5.5%) presentano probabilita' di intrusione superiore a 0.5.
#>
#> ## Purezza delle US
#>
#> Su 9 US: 0 pure (>90%), 0 miste (70-90%), 9 palinsesto (<70%).
#>
#> ---
#> *Rapporto generato da palimpsestr 0.18.0*

```

9 Database Import

Import data from SQLite or PostgreSQL:

```
# SQLite
data <- read_db(
  con = DBI::dbConnect(RSQLite::SQLite(), "excavation.sqlite"),
  table = "finds"
)

# PostgreSQL (PyArchInit)
data <- read_db(
  con = DBI::dbConnect(RPostgres::Postgres(), dbname = "pyarchinit", host = "localhost"),
  table = "finds_table"
)
```

10 Phase Reordering

Mixture models suffer from *label switching*: phase numbers are arbitrary across runs. Use `reorder_phases()` to ensure phase 1 = deepest (oldest) and phase K = shallowest (most recent):

```
fit <- reorder_phases(fit)
table(fit$phase)
#>
#> 1 2 3
#> 67 67 66
```

11 Bootstrap Confidence Intervals

Assess uncertainty on key metrics via bootstrap resampling:

```
bs <- bootstrap_sef(fit, n_boot = 50, verbose = FALSE)
print(bs)
#>      statistic      estimate      lower      upper      se
#> 1          pdi 9.999945e-01 9.998558e-01 1.000000e+00 6.087534e-05
#> 2 mean_entropy 6.095726e-06 6.806146e-11 1.583963e-04 6.687840e-05
#> 3 mean_energy 2.008381e+00 1.880952e+00 2.022657e+00 4.105364e-02
#> 4          loglik -1.177707e+03 -1.190941e+03 -1.113264e+03 2.306197e+01
```

Returns a data.frame with columns `metric`, `mean`, `se`, `ci_lower`, `ci_upper` for PDI, mean entropy, mean energy, loglik, and optionally ARI (if `true_labels` is provided).

Parameters:

	Parameter	Default	Description
<code>n_boot</code>	100		Number of bootstrap replicates
<code>conf</code>	0.95		Confidence level for intervals
<code>true_labels</code>	NULL		Known true phases for ARI confidence interval
<code>verbose</code>	TRUE		Print progress messages

12 Cross-Validation

Evaluate model fit via k-fold cross-validation:

```

cv <- cv_sef(moderate, k_values = 3, n_folds = 3, seed = 42)
print(cv)
#>   k fold train_loglik test_loglik train_pdi
#> 1 3     1    -777.3906   -412.3507 0.9999997
#> 2 3     2    -761.4142   -439.0358 1.0000000
#> 3 3     3    -792.4940   -386.4890 0.9999999

```

Use with multiple K values to find the best:

```

cv <- cv_sef(data, k_values = 2:5, n_folds = 5, seed = 42)
print(cv)  # shows mean_test_loglik per k

```

13 Data-Driven Weight Optimisation

Instead of manually tuning SEI weights, use `optimize_weights()` to find the best configuration via grid search:

```

opt <- optimize_weights(data, k = 3, n_folds = 3, seed = 42)
print(opt$best_weights)

# Re-fit with optimal weights
fit <- fit_sef(data, k = 3, weights = opt$best_weights)

```

14 Sparse SEI for Large Datasets

For datasets with $n > 1000$, the full $n \times n$ SEI matrix may be too large. Use `sei_sparse()` which automatically applies a distance threshold:

```

# Automatic sparse mode
sei <- sei_sparse(data, max_dist = 5.0)

# Or use sei_matrix() directly with max_dist
sei <- sei_matrix(data, max_dist = 5.0)

```

Pairs beyond `max_dist` Euclidean distance receive SEI = 0, dramatically reducing memory and computation time.

15 Complete Workflow Example

```

library(palimpsestr)

# 1. Load data
data <- read.csv("site_data.csv")

# 2. Select optimal K
ck <- compare_k(data, k_values = 2:6, context = "context",
                 tafonomy = "taf_score", n_init = 3)
gg_compare_k(ck)
best_k <- ck$k[which.min(ck$bic)]

# 3. Fit with optimal K and multiple inits
fit <- fit_sef(data, k = best_k, context = "context",
               tafonomy = "taf_score", n_init = 5, seed = 42)

```

```

# 4. Inspect results
print(fit)
gg_phasefield(fit)
gg_entropy(fit)
gg_convergence(fit)
gg_phase_profile(fit)

# 5. Intrusion detection
gg_intrusions(fit, top_n = 10)

# 6. Harris validation
validate_phases_harris(fit)

# 7. Export
export_results(fit, dir = "results/")
report_sf(fit, lang = "en")

# 8. GIS
sf_pts <- as_sf_phase(fit, crs = 32632L)
sf::st_write(sf_pts, "results.gpkg", layer = "phases")

```

16 Real-Data Example: Villa Romana

The `villa_romana` dataset is derived from the multi-period Roman villa of Poggio Gramignano (Lugnano in Teverina, Italy): 615 inventoried finds across 54 stratigraphic units and 17 material classes, spanning the Pre-Roman to Late Antique periods, with excavator-assigned taphonomic scores. Note that coordinates and dating are recorded at stratigraphic-unit level (see `recommend_setup()`), so the model resolves phases at the unit and macro-event level.

16.1 Load and inspect

```

data(villa_romana)
cat("Finds:", nrow(villa_romana), "| Contexts:", length(unique(villa_romana$context)),
    "| Classes:", length(unique(villa_romana$class)), "\n")
#> Finds: 615 | Contexts: 54 | Classes: 17
cat("Date range:", min(villa_romana$date_min), "to", max(villa_romana$date_max), "\n")
#> Date range: -650 to 550
table(villa_romana$class)
#>
#>      Campionature      Intonaco Dipinto Materiale Da Costruzione
#>              5              1              38
#>      Materiale Lapideo Pietre Dure-Gemme  Reperti Antropologici
#>              1              1              1
#> Reperti Archeozoologici Reperto Anforaceo      Reperto Ceramico
#>              7              76             417
#>      Reperto Faunistico Reperto Lapideo      Reperto Metallico
#>              1              6              19
#>      Reperto Numismatico Reperto Vitreo Resti Di Pavimentazioni
#>             10             25              1
#>      Scarti Di Produzione      Stucco
#>              5              1

```

16.2 Fit with taphonomic weighting

```
fit_vr <- fit_sef(villa_romana, k = 4,
                 tafonomy = "taf_score",
                 context   = "context",
                 n_init    = 5, seed = 42)

print(fit_vr)
#> <sef_fit>
#> Observations: 615
#> Estimated phases: 4
#> Mean entropy: 0.000
#> Mean local SEI: 1175.378
#> Mean energy: 0.882
#> LogLik: 3536.812
#> BIC: -6386.510
#> ICL: -6386.524
#> PDI: 1.000
#> Converged: yes
#> Class model: multinomial
#> Initialisations: 5
```

16.3 Diagnostics

```
cat("PDI:", pdi(fit_vr), "\n")
#> PDI: 0.9999916
sef_summary(fit_vr)
#> $n
#> [1] 615
#>
#> $k
#> [1] 4
#>
#> $pdi
#> [1] 0.9999916
#>
#> $mean_entropy
#> [1] 1.159321e-05
#>
#> $mean_energy
#> [1] 0.8821964
#>
#> $loglik
#> [1] 3536.812
#>
#> $bic
#> [1] -6386.51
#>
#> $phase_count
#>
#>   1   2   3   4
#> 436 48 121 10
```

16.4 Validation (on simulated data)

Validation with ARI and confusion matrix requires known true phases. We demonstrate with `demo_moderate`:

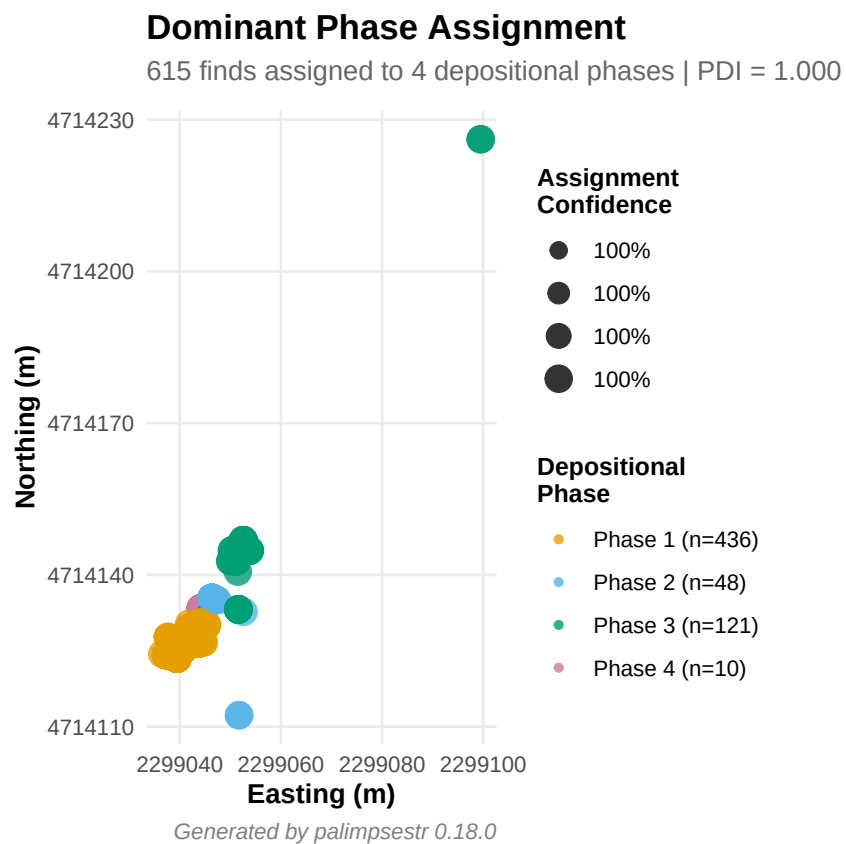
```
data(demo_moderate)
fit_demo <- fit_sef(demo_moderate, k = 3, seed = 42)
adjusted_rand_index(fit_demo, demo_moderate$true_phase)
#> [1] 1
confusion_matrix(fit_demo, demo_moderate$true_phase)
#>           true
#> estimated  2  1  3
#>           1 67  0  0
#>           2  0 67  0
#>           3  0  0 66
```

16.5 Intrusion detection

```
di_vr <- detect_intrusions(fit_vr)
suspects_vr <- di_vr[di_vr$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects_vr), "out of", nrow(villa_romana), "\n")
#> Suspected intrusions: 3 out of 615
```

16.6 Visualisation

```
gg_phasefield(fit_vr)
```

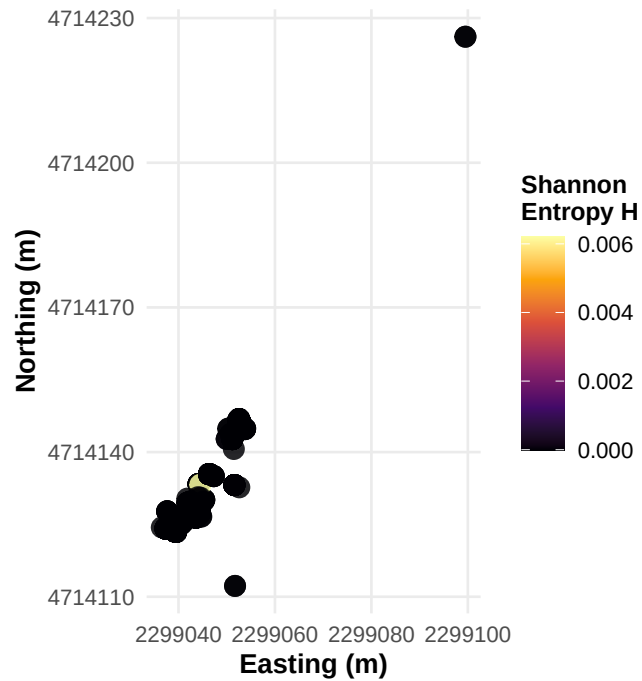


```
gg_entropy(fit_vr)
```

Spatial Entropy Map

Entropy measures uncertainty in phase assignment

Dark = confident assignment | Bright = ambiguous (possible mixing ;



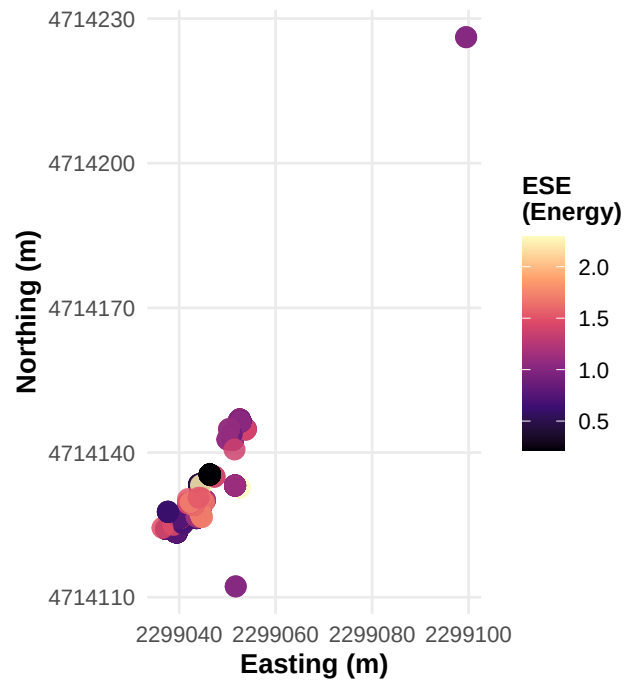
Generated by palimpsestr 0.18.0

```
gg_energy(fit_vr)
```

Excavation Stratigraphic Energy (ESE)

Energy measures local depositional disruption

Dark = stable deposit | Bright = mixing zone (bioturbation, redeposit



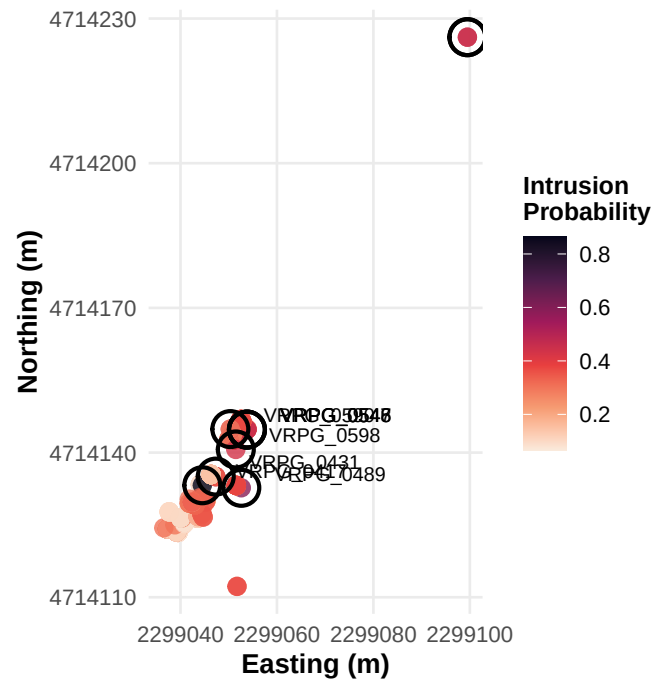
Generated by palimpsestr 0.18.0

```
gg_intrusions(fit_vr, top_n = 10)
```

Intrusion Detection Map

3 finds flagged (prob > 0.5) | Circled: top 10 suspects

Score combines high entropy + high energy + low network connectiv

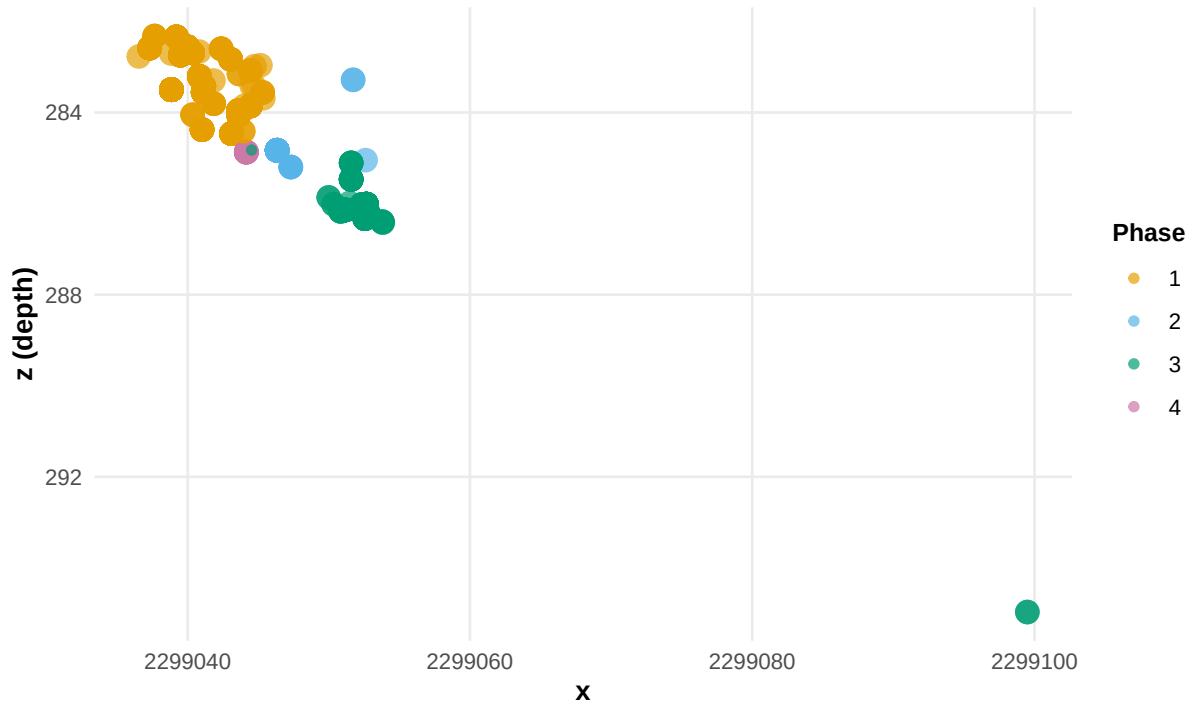


Generated by palimpsestr 0.18.0

```
gg_phase_profile(fit_vr)
```

Vertical Phase Profile

Depth (z) vs horizontal position, coloured by phase



Generated by palimpsestr 0.18.0

16.7 Per-US summary

`us_summary_table(fit_vr)`

```
#>   context n_finds dominant_phase purity mean_entropy mean_energy
#> 1  US_102     13             1      1 1.348904e-12  0.9116826
#> 2  US_104     93             1      1 8.006380e-18  0.7710822
#> 3  US_105     19             1      1 3.229626e-20  0.6686666
#> 4  US_108      3             1      1 3.517656e-14  0.8569123
#> 5  US_111     25             1      1 3.385322e-24  0.9672565
#> 6  US_114     22             1      1 6.073610e-15  0.8245524
#> 7  US_116      6             1      1 1.609169e-15  1.0170747
#> 8  US_117     27             1      1 7.201187e-23  0.9095915
#> 9  US_118     16             1      1 4.617782e-11  0.9341534
#> 10 US_133      1             1      1 3.567652e-21  1.4766197
#> 11 US_134     41             1      1 1.393958e-16  0.7658666
#> 12 US_136     33             1      1 9.487312e-10  0.9885305
#> 13 US_137     15             1      1 2.139375e-10  1.1168077
#> 14 US_144     18             1      1 3.009715e-10  1.1194293
#> 15 US_148      1             1      1 3.903770e-19  0.7056760
#> 16 US_149     15             1      1 1.766526e-19  0.6630704
#> 17 US_150     48             1      1 1.401461e-20  0.8866959
#> 18 US_151     10             4      1 2.189355e-10  0.5242270
#> 19 US_157      3             1      1 1.546477e-09  1.1795692
#> 20 US_158      1             1      1 6.542247e-10  1.1824122
```

```

#> 21 US_161      4      1      1 1.085694e-13 1.4024354
#> 22 US_166      2      1      1 5.154336e-11 1.3664971
#> 23 US_173      1      3      1 6.215833e-03 2.2285077
#> 24 US_180      1      1      1 4.676603e-25 1.4738468
#> 25 US_183      5      1      1 1.238610e-24 0.6177584
#> 26 US_189      2      1      1 4.967867e-14 1.3148516
#> 27 US_197      3      1      1 6.187898e-13 1.6754428
#> 28 US_201      1      1      1 2.153991e-14 1.5198388
#> 29 US_204      4      2      1 8.614016e-11 1.6094968
#> 30 US_209      1      1      1 2.767693e-11 1.4576088
#> 31 US_21      2      2      1 4.006853e-58 1.0000000
#> 32 US_210     41      2      1 0.000000e+00 0.2834099
#> 33 US_211      1      1      1 1.431351e-12 1.6650720
#> 34 US_215      1      1      1 3.888527e-15 1.6986328
#> 35 US_220      1      1      1 6.717062e-14 1.6964680
#> 36 US_258      8      1      1 5.487815e-15 1.5910054
#> 37 US_276      1      2      1 3.671074e-19 2.2891247
#> 38 US_287     25      3      1 1.166596e-07 0.9416951
#> 39 US_296     10      3      1 4.292317e-10 0.9686296
#> 40 US_297     17      3      1 1.371370e-09 0.8483098
#> 41 US_299      1      3      1 2.312728e-11 1.0080910
#> 42 US_302      2      3      1 0.000000e+00 1.0000000
#> 43 US_303      1      3      1 2.291997e-11 1.0066694
#> 44 US_304      3      3      1 3.588579e-12 1.3775978
#> 45 US_306      5      3      1 4.949377e-09 1.0256186
#> 46 US_307     30      3      1 3.959752e-09 0.9129552
#> 47 US_308      4      3      1 1.228439e-07 1.0316612
#> 48 US_315      2      3      1 9.861311e-06 1.1318639
#> 49 US_316      2      3      1 8.074350e-07 1.1944883
#> 50 US_319      6      3      1 1.059890e-04 0.9671551
#> 51 US_324      1      3      1 5.366125e-09 1.3699999
#> 52 US_72       5      1      1 9.967366e-17 1.4994895
#> 53 US_76      11      3      1 2.300811e-05 1.0012723
#> 54 US_89       1      1      1 6.263975e-11 1.5259711
#>      mean_local_sei n_intrusions
#> 1      1475.67927      0
#> 2      1454.47240      0
#> 3      1462.31198      0
#> 4      1489.49232      0
#> 5      1296.11532      0
#> 6      1440.29144      0
#> 7      1337.52512      0
#> 8      1339.40955      0
#> 9      1366.51078      0
#> 10     1092.23363      0
#> 11     1459.61396      0
#> 12     1376.46698      0
#> 13     1198.39339      0
#> 14     1236.64246      0
#> 15     1481.23740      0
#> 16     1440.03425      0
#> 17     1407.46372      0
#> 18     1273.79366      0

```

```

#> 19      1428.85433      0
#> 20      1076.40240      0
#> 21      1060.04979      0
#> 22      1027.26673      0
#> 23        606.90952      1
#> 24      1066.93451      0
#> 25      1362.36933      0
#> 26      1270.42593      0
#> 27      1041.68052      0
#> 28      1102.20707      0
#> 29        710.52499      1
#> 30      1091.58828      0
#> 31        592.93607      0
#> 32        854.27737      0
#> 33      1044.09577      0
#> 34      1034.54278      0
#> 35        998.15146      0
#> 36      1061.88830      0
#> 37        368.35027      1
#> 38        559.45797      0
#> 39        604.88564      0
#> 40        654.84219      0
#> 41        781.72219      0
#> 42         60.16121      0
#> 43        781.46404      0
#> 44        383.27231      0
#> 45        497.67760      0
#> 46        579.06540      0
#> 47        523.70113      0
#> 48        642.66694      0
#> 49        615.68993      0
#> 50        720.36507      0
#> 51        413.37717      0
#> 52      1204.92622      0
#> 53        618.10970      0
#> 54      1010.33897      0

```

16.8 Interpretive report

```

report_sef(fit_vr, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with K = 4 phases was fitted to a dataset of 615 fi
#>
#> The Palimpsest Dissolution Index (PDI) is 1.000, indicating excellent phase separation. Mean
#>
#> Model diagnostics: LogLik = 3536.8, BIC = -6386.5.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (436 finds, 70.9%)

```

```

#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Materiale Da Costruzione (20), Reperto Anforaceo (62), Reperto Ceramico (319)
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.912
#> - **Contexts**: US_102, US_104, US_105, US_108, US_111, US_114, ... (33 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (48 finds, 7.8%)
#>
#> - **Dating**: average 374 BCE (range: -650 to 550)
#> - **Material classes**: Materiale Da Costruzione (3), Reperti Archeozoologici (1), Reperto Anforaceo
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.466
#> - **Contexts**: US_204, US_21, US_210, US_276
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (121 finds, 19.7%)
#>
#> - **Dating**: average 372 CE (range: 180 to 425)
#> - **Material classes**: Campionature (5), Intonaco Dipinto (1), Materiale Da Costruzione (15), Mater
#> - **Mean entropy**: 0.0001; **Mean energy**: 0.971
#> - **Contexts**: US_173, US_287, US_296, US_297, US_299, US_302, ... (16 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 4 (10 finds, 1.6%)
#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Reperto Anforaceo (1), Reperto Ceramico (9)
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.524
#> - **Contexts**: US_151
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **3 finds** (0.5%) have an intrusion probability above 0.5, flagging them as potentially displaced o
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | VRPG_0417 | US_173 | 0.864 |
#> | VRPG_0489 | US_276 | 0.593 |
#> | VRPG_0431 | US_204 | 0.578 |
#>
#> ## Stratigraphic Unit Purity
#>
#> Of **54 stratigraphic units**: **54 pure** (>90%), **0 mixed** (70-90%), **0 palimpsest** (<70%).
#>
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---

```

17 Function Reference

17.1 Core

Function	Description
<code>fit_sef()</code>	Fit the SEF model
<code>compare_k()</code>	Compare multiple K values
<code>archaeo_sim()</code>	Simulate archaeological data
<code>reorder_phases()</code>	Relabel phases by mean depth (deepest = phase 1)

17.2 Diagnostics

Function	Description
<code>pdi()</code>	Palimpsestr Dissolution Index
<code>sef_summary()</code>	Compact model summary
<code>as_phase_table()</code>	Phase probability table
<code>predict_phase()</code>	Alias for <code>as_phase_table()</code>
<code>phase_diagnostic_table()</code>	Full diagnostic table
<code>detect_intrusions()</code>	Intrusion detection

17.3 Validation

Function	Description
<code>adjusted_rand_index()</code>	Adjusted Rand Index
<code>confusion_matrix()</code>	Confusion matrix (auto-reordered)

17.4 Export

Function	Description
<code>us_summary_table()</code>	Per-US summary statistics
<code>phase_transition_matrix()</code>	Vertical transition counts
<code>export_results()</code>	Export all results to CSV

17.5 Harris Matrix

Function	Description
<code>harris_from_contexts()</code>	Auto-generate penalty from depths
<code>read_harris()</code>	Import Harris Matrix from CSV
<code>validate_phases_harris()</code>	Check phase-stratigraphy consistency

17.6 Robustness & Sensitivity

Function	Description
<code>bootstrap_sef()</code>	Bootstrap confidence intervals on PDI, entropy, ARI
<code>cv_sef()</code>	K-fold cross-validation for model selection
<code>optimize_weights()</code>	Data-driven SEI weight optimisation via grid search

17.7 SEI / ESE

Function	Description
<code>sei_matrix()</code>	Stratigraphic Entanglement Index matrix
<code>sei_sparse()</code>	Sparse SEI for large datasets (n > 1000)
<code>local_sei()</code>	Row sums of SEI matrix
<code>ese()</code>	Excavation Stratigraphic Energy

17.8 Plots (ggplot2)

Function	Description
<code>gg_phasefield()</code>	Phase assignment map
<code>gg_entropy()</code>	Entropy map
<code>gg_energy()</code>	ESE energy map
<code>gg_intrusions()</code>	Intrusion detection map
<code>gg_compare_k()</code>	K selection diagnostics
<code>gg_convergence()</code>	EM convergence trace
<code>gg_phase_profile()</code>	Vertical depth profile
<code>gg_confusion()</code>	Confusion matrix heatmap
<code>gg_cv()</code>	Cross-validation diagnostic plot
<code>gg_bootstrap()</code>	Bootstrap confidence interval plot
<code>gg_weights()</code>	Weight sensitivity heatmap
<code>gg_map()</code>	Overlay on excavation plan
<code>as_plotly()</code>	Convert any gg_* to interactive

17.9 Plots (base R)

Function	Description
<code>plot_phasefield()</code>	Phase map
<code>plot_entropy()</code>	Entropy map
<code>plot_energy()</code>	Energy map
<code>plot_sei_profile()</code>	SEI profile

17.10 GIS

Function	Description
<code>as_sf_phase()</code>	Export as sf point layer
<code>as_sf_links()</code>	Export SEI links as sf lines
<code>load_geometries()</code>	Load excavation plan geometries

17.11 Other

Function	Description
<code>report_sef()</code>	Interpretive report (en/it)
<code>read_db()</code>	Import from SQLite/PostgreSQL